

Computer Networks

A computer network is a group of computer systems and other computing hardware and network devices that are linked together through communication channels to facilitate communication and resource-sharing among a wide range of users.

Benefits of Computer Networks

1. Resource Sharing

If there are 5 computers in an office allocated to different employees and all the employees have been given a separate printer attached to their computer.

Now it is felt that these employees do not an individual printer but can manage with just one printer.

- Because it is increasing the cost, i.e. 5 printers have to be bought in place of one which increases the costs.
- Because 5 printers encroach 5 times the space in the office.

Solution: Computer Network.

All the 5 computers can be connected to 1 printer on the network to save cost and space.

2. Information Sharing

Computer Networks are used to share information amongst each other. For example: Websites. If an organisation or a company wants to share some information with a large number of people, website is a good solution and it can be accessed via internet. Universities and organisations have their own internal networks for the purpose of information sharing.

3. Fast transfer rate

When we wish to send some data from one place to another and if we send it via a storage medium i.e. a hard drive, we will have to copy data to the hard drive and send it. It will take a long time to reach.

In the other case if the two locations are connected through a network the same data can be transferred within no time.

4. Saving of time

Networks also help in saving time. Suppose we wish to withdraw some money from the bank we now don't have to travel to the bank, we can now withdraw the money from the ATM machine nearby since it's on the network.

5. Increased Storage Capacity

As there is more than one computer on a network which can easily share files, the issue of storage capacity gets resolved to a great extent. A standalone computer might fall short of storage memory,

but when many computers are on a network, memory of different computers can be used in such case. One can also design a storage server on the network in order to have a huge storage capacity.

6. Increased Cost Efficiency

There are many softwares available in the market which are costly and take time for installation.

Computer networks resolve this issue as the software can be stored or installed on a system or a server and can be used by the different workstations

Disadvantages of Computer Networks

1. Security Issues

One of the major drawbacks of computer networks is the security issues involved. If a computer is a standalone, physical access becomes necessary for any kind of data theft. However, if a computer is on a network, a computer hacker can get unauthorized access by using different tools.

In case of big organizations, various network security softwares are used to prevent the theft of any confidential and classified data.

2. Rapid Spread of Computer Viruses

If any computer system in a network gets affected by computer virus, there is a possible threat of other systems getting affected too. Viruses get spread on a network easily because of the interconnectivity of workstations. Such spread can be dangerous if the computers have important database which can get corrupted by the virus.

3. Expensive Set Up

The initial set up cost of a computer network can be high depending on the number of computers to be connected. Costly devices like routers, switches, hubs, etc., can add up to the bills of a person trying to install a computer network. He will also have to buy NICs (Network Interface Cards) for each of the workstations, in case they are not inbuilt.

4. Dependency on the Main File Server

In case the main File Server of a computer network breaks down, the system becomes useless. In case of big networks, the File Server should be a powerful computer, which often makes it expensive.

Classification of Computer Networks

Computer networks can be classified in the following ways:-

- 1) Based on Size
- 2) Based on Topology
- 3) Based on Media Used
- 4) Based on Interaction

1) Based of Size

a) LAN: LANs (Local Area Networks) are a network infrastructure that spans a small geographical area like a building or a campus. Specific features of LANs include:

- LANs interconnect end devices in a limited area such as a home, school, office building, or campus.
- A LAN is usually administered by a single organization or individual. The administrative control that governs the security and access control policies are enforced on the network level.
- LANs provide high speed bandwidth to internal end devices and intermediary devices.

b) MAN (Metropolitan Area Network):

A network infrastructure that spans a physical area larger than a LAN but smaller than a WAN (e.g., a city). MANs are typically operated by a single entity such as a large organization.

c) WAN (Wide Area Network):

WANs are a network infrastructure that spans a wide geographical area.

Specific features of WANs include:

- WANs interconnect LANs over wide geographical areas such as between cities, states, provinces, countries, or continents.
- WANs are usually administered by multiple service providers.
- WANs typically provide slower speed links between LANs.

2) Based on Topology

a) Star Topology

- Star Topology is named so because the computers when connected in this configuration look like a star.
- All the computers are connected to one hub/switch.
- No computer is connected to any other computer other than the hub/switch.
- If any computer wants to communicate with any other computer has to go through the hub/switch.
- This kind of topology is useful where there is requirement of hub/switch.

Figure showing a star topology

Ring Topology

- Ring Topology is named so because the computers in this topology are connected in a ring.
- In a Ring the communication is only in one direction.
- Suppose if the direction of communication is towards right, any computer can communicate to the computer immediately to its right.
- But it cannot communicate to the computer immediately to its left.
- The message has to go through the ring towards Right and travel the whole ring to reach the left computer.

Figure of a ring topology

BUS Topology

- In this configuration all the computers are connected to a common backbone called the BUS.
- All the computers can transmit directly to all other computers.
- The only problem is that if two or more than two computers transmit at the same time, then there can be a data collision on the network.
- But there is a solution to this problem called CSMA/CD (Carrier Sense Multiple Access/Collision Detection). This technology enables all the computers to transmit on the same common BUS.

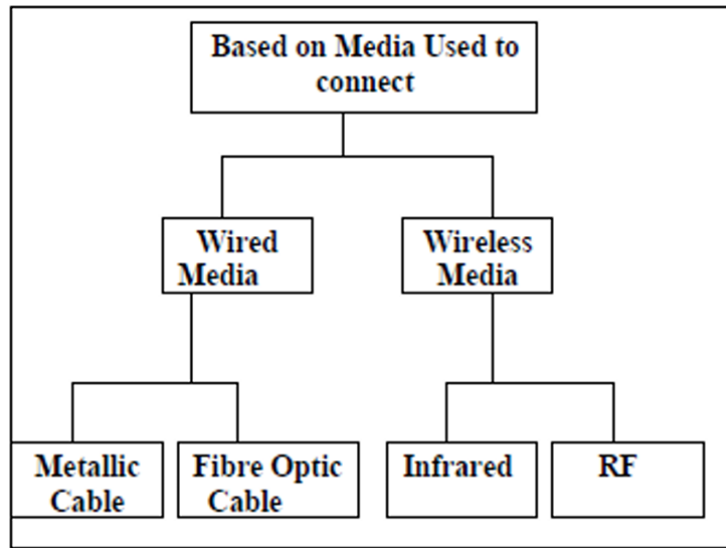
Figure of BUS Topology

MESH Topology

- This is an ideal configuration though not practically implementable because we can't physically connect each computer on a network to every other computer on the network.
- It will create very large number of physical cable connections.

Figure of MESH Topology

3) Based on Media



The networks can be either wired or wireless networks.

a) The wired networks can be connected either using metallic cables like copper cables, Coaxial Cables, UTP Cables etc. Or the network can be connected using Fibre Optic Cables.

b) The wireless networks can be connected using either Infra-Red medium, microwave, Radio-Frequency, satellite, etc.

4) Based on Interaction

a) Peer-to-Peer Network

In a peer to peer network, a number of workstations (or clients) are connected together for sharing devices, information or data. All the workstations (clients) are considered equal. Any one computer can act as client or server at any instance. This network is ideal for small networks where there is no need for dedicated servers, like home networks, small business networks, or retail shops.

Advantages of a Peer-to-Peer network

- It is easy to install and so is the configuration of computers on this network,
- All the resources and contents are shared by all the peers, unlike server-client architecture where Server shares all the contents and resources.
- It is more reliable as central dependency is eliminated. Failure of one peer doesn't affect the functioning of other peers.

- There is no need for full-time System Administrator. Every user is the administrator of his machine. User can control their shared resources.
- The over-all cost of building and maintaining this type of network is comparatively very less.

Disadvantages of a Peer-to-Peer network

- In this network, the whole system is decentralized thus it is difficult to administer. That is one person cannot determine the whole accessibility setting of whole network.
- Security in this system is very less. Viruses, spywares, trojans, etc malwares can easily be transmitted over this architecture.
- Data recovery or backup is very difficult. Each computer should have its own back-up system
- Lot of movies, music and other copyrighted files are transferred using this type of file transfer.

Client-Server Network

In a Client-Server network, one or more powerful computers (servers) provide the different computer network services and all other users of computer network (clients) access those services to perform user's tasks.

- In such networks, there exists a central controller called server. A server is a specialized computer that controls the network resources and provides services to other computers in the network.
- All other computers in the network are called clients. A client computer receives the requested services from a server.
- A server performs all the major operations like security and network management.
- All the clients communicate with each other via centralized server. If client 1 wants to send data to client 2, it first sends request to server to seek permission for it. The server then sends a signal to client 1 allowing it to initiate the communication.
- A server is also responsible for managing all the network resources such as files, directories, applications & shared devices like printer etc.
- If any of the clients wants to access these services, it first seeks permission from the server by sending a

Note: Most Local Area Networks are based on client server relationship.

Advantages of Client Server Networks

- Centralized back up is possible.
- Use of dedicated server improves the performance of whole system request.
- Security is better in these networks as all the shared resources are centrally administered.
- Use of dedicated servers also increases the speed of sharing resources.
- There is no limit to the number of computers that can be connected.

Disadvantages of Client Server Networks

- It requires specialized servers with large memory and secondary storage. This leads to increase in the cost.

- The cost of network operating system that manages the various clients is also high.
- It requires dedicated network administrator.